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**Subject:** Machine Learning Laboratory (Semester V)

**Project Title:** 92 Workforce Demand Planning Using Machine Learning

**GitHub Link:** <https://github.com/Shweta-Tech-creator/WorkForce.git>

**Live App Link:** <https://workforce-demand-planning.streamlit.app>

## Workforce Demand Planning

### Abstract

Workforce planning is important for every organization. Too few employees can increase workload, while too many can lead to unnecessary expenses.

This project uses machine learning to classify workforce demand as **Low, Medium, or High** using the IBM HR Analytics dataset of 1,470 employee records. Four models were trained and evaluated, with **Random Forest achieving the highest accuracy of 90.82%** and an F1-score of 0.9064.

An interactive **Streamlit dashboard** was also developed to explore employee data, estimate demand, and support workforce planning decisions.

## 1. Introduction & Problem Statement

### 1.1 Why Workforce Planning Matters

Workforce planning helps organizations ensure they have the right number of employees at the right time. When companies only hire after employees resign or workloads increase, teams may become overworked and projects can be delayed. Planning ahead helps HR teams prepare for these situations and manage workloads effectively.

### 1.2 Problem Statement

The aim of this project is to estimate workforce demand for different roles or departments by analyzing factors such as workload, employee satisfaction, performance, and attrition.

### 1.3 The 3 Output Classes

The system classifies workforce demand into three simple categories to make the results easier to understand and use.

- **Low Demand:** The current workforce appears sufficient, with no immediate need for additional hiring.
- **Medium Demand:** Some additional staffing or replacement planning may be required.
- **High Demand:** Workload or attrition indicators suggest a need to consider hiring or redistributing tasks.

## 2. Dataset Overview

For this project, I used the **IBM HR Analytics Employee Attrition & Performance** dataset from Kaggle. It contains employee information that helps analyze workload, job satisfaction, performance, and attrition.

### 2.1 Dataset Summary

| Parameter        | Details                                           |
|------------------|---------------------------------------------------|
| Dataset Name     | IBM HR Analytics Employee Attrition & Performance |
| Source           | Kaggle                                            |
| Total Employees  | 1,470                                             |
| Initial Columns  | 35                                                |
| Cleaned Features | 31                                                |
| Encoded Features | 45                                                |
| Target Variable  | Workforce_Demand (Low, Medium, High)              |

### 2.2 Key Features in the Dataset

The dataset includes different factors that help understand employee and workplace conditions:

- **Workload:** Overtime, business travel, distance from home, and work-life balance.
- **Career & Tenure:** Total working years, years at company, current role, and last promotion.
- **Performance & Satisfaction:** Performance rating, job involvement, and job satisfaction.
- **Job Details:** Department, job role, job level, and monthly income.
- **Attrition:** Indicates whether an employee has left the organization.

These features were used to prepare the data for training and evaluating the machine learning models.

### 3. Data Cleaning & Preprocessing

Before training the models, I cleaned and prepared the dataset to make it suitable for machine learning.

1. **Missing Values and Duplicates:** I checked the dataset and found no missing values or duplicate records.
2. **Removing Unnecessary Columns:** I removed four columns: EmployeeCount, StandardHours, Over18, and EmployeeNumber, as they were not useful for prediction.
3. **Categorical Encoding:** I used One-Hot Encoding with `pd.get_dummies(drop_first=True)` to convert text-based data into numerical values, resulting in **45 features**.
4. **Train-Test Split:** I divided the data into **80% training (1,176 records)** and **20% testing (294 records)**, using stratification to maintain the demand-class proportions.
5. **Feature Scaling:** I applied StandardScaler to numerical features so that models such as Logistic Regression and KNN could work with features on a comparable scale.

### 4. Creating the Workforce Demand Target

Since the dataset had no demand label, I created a **Workforce Demand Score** using factors such as overtime, attrition, work-life balance, performance, and job satisfaction.

Based on the score, employees were grouped into three categories:

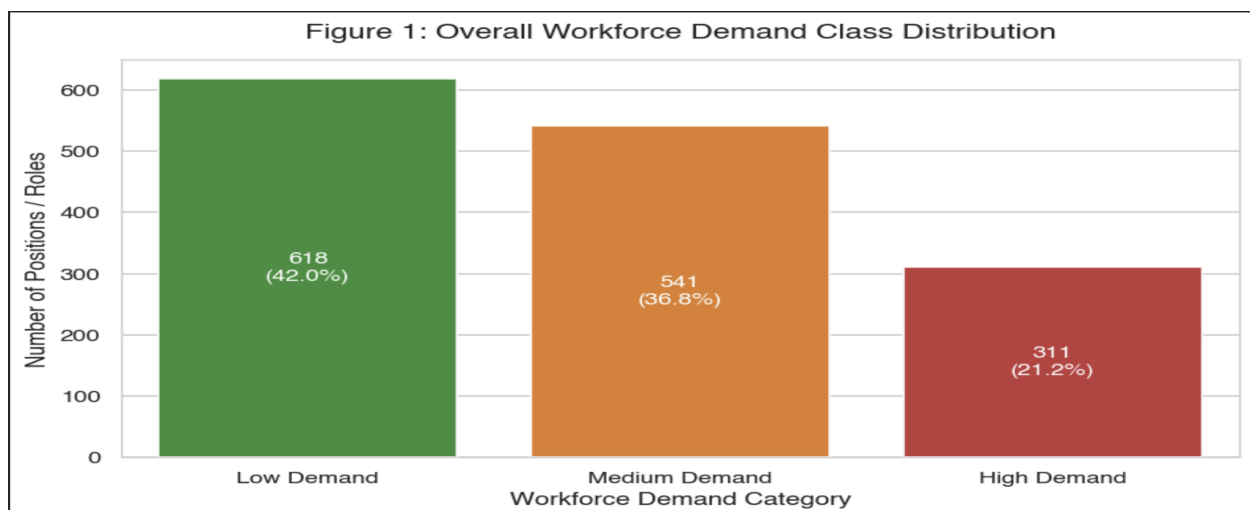
| Category      | Score Range | Records | Percentage |
|---------------|-------------|---------|------------|
| Low Demand    | $\leq 1.5$  | 618     | 42.04%     |
| Medium Demand | 2.0–3.5     | 541     | 36.80%     |
| High Demand   | $\geq 4.0$  | 311     | 21.16%     |

These categories help identify potential staffing needs and support workforce planning.

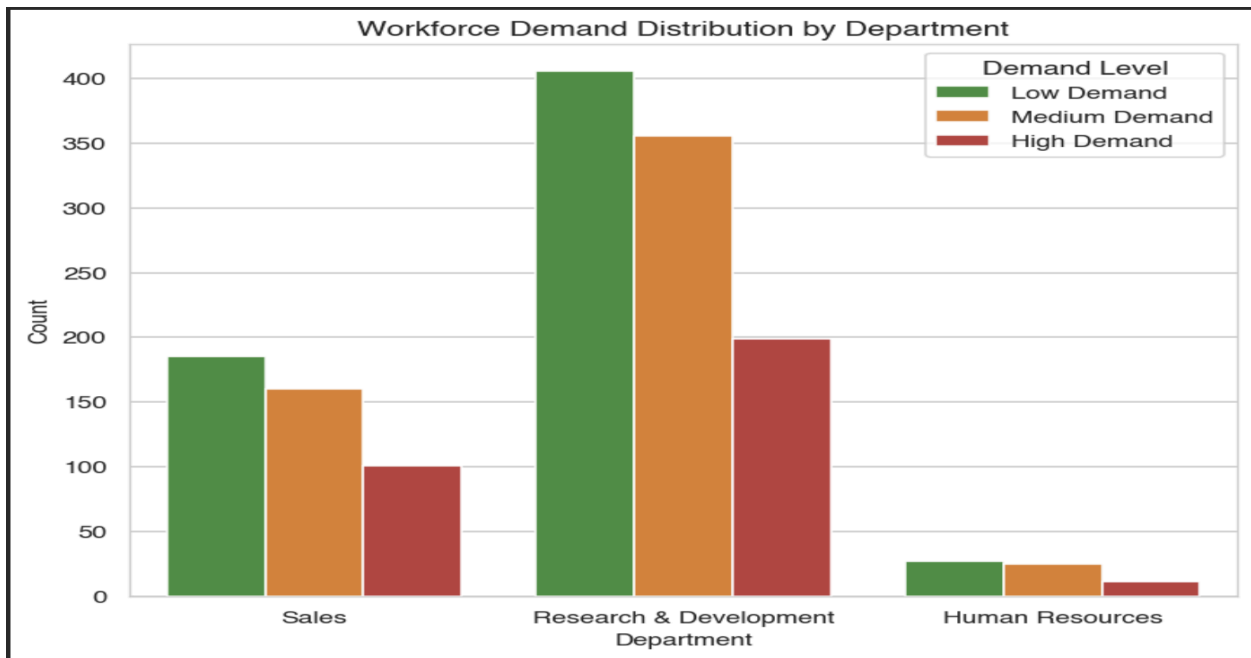
## 5. Exploratory Data Analysis (EDA) & Key Findings

### 5.1 Visualizations & Observations

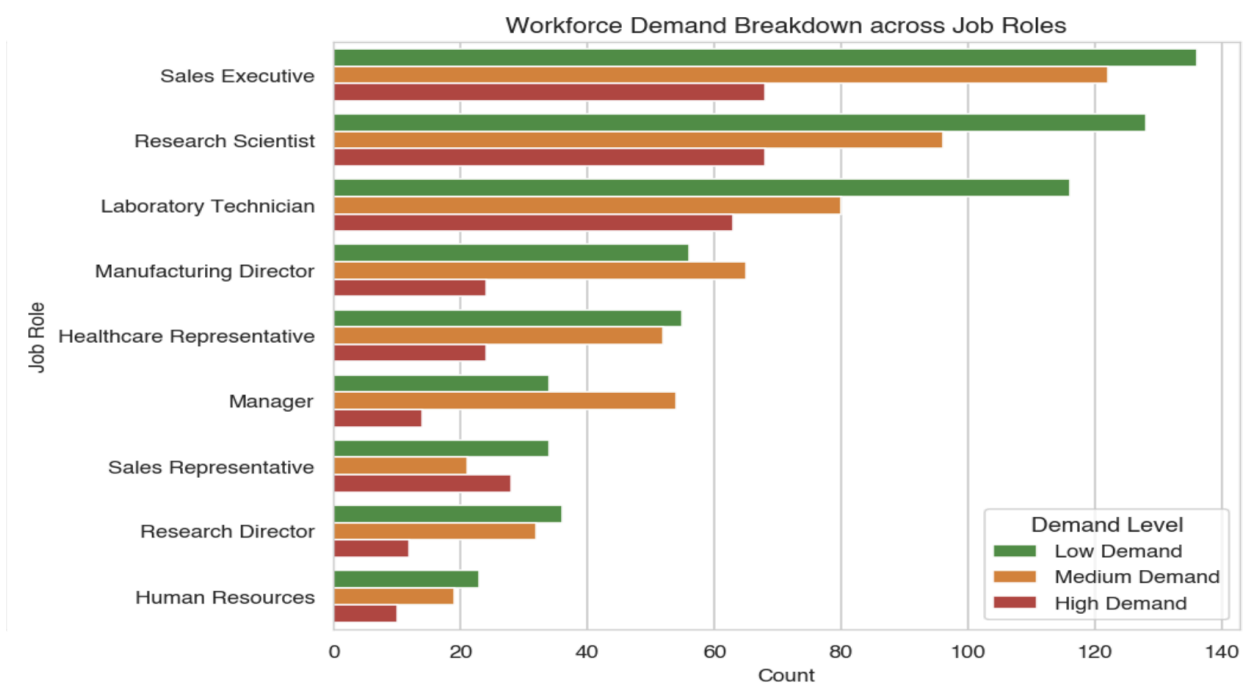
**1. Target Class Distribution :** The dataset contains three demand categories: Low, Medium, and High. Low Demand has the highest number of records, followed by Medium and High Demand.



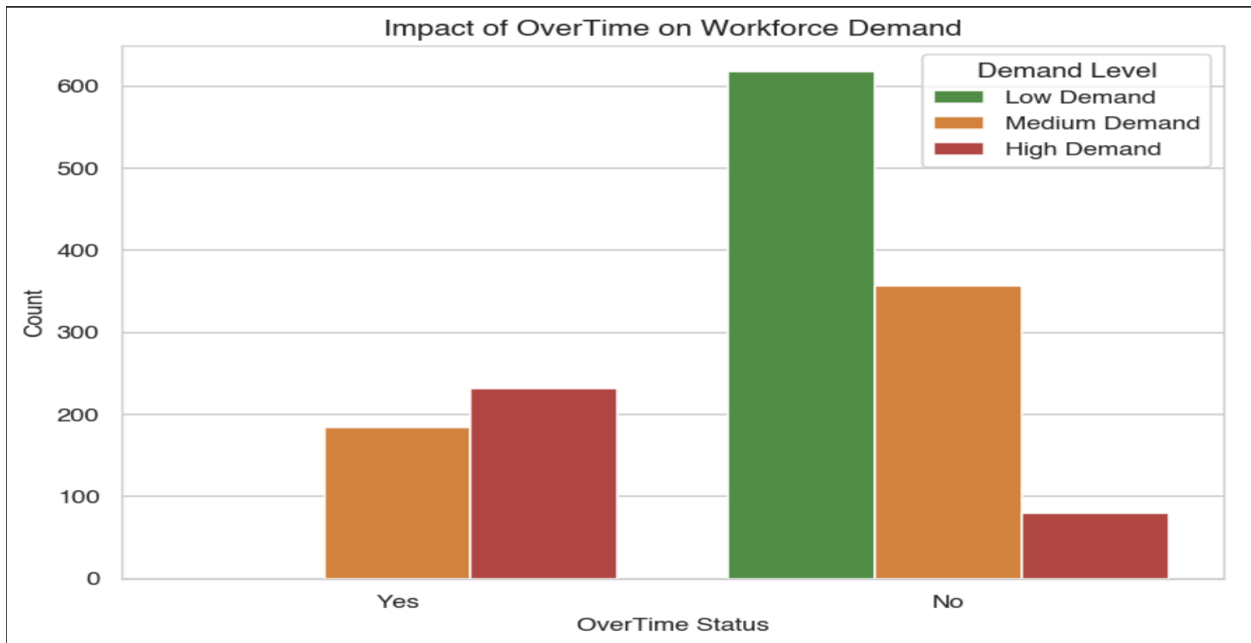
**2. Demand Breakdown by Department:** The chart compares workforce demand across departments and helps identify differences in demand categories between teams.



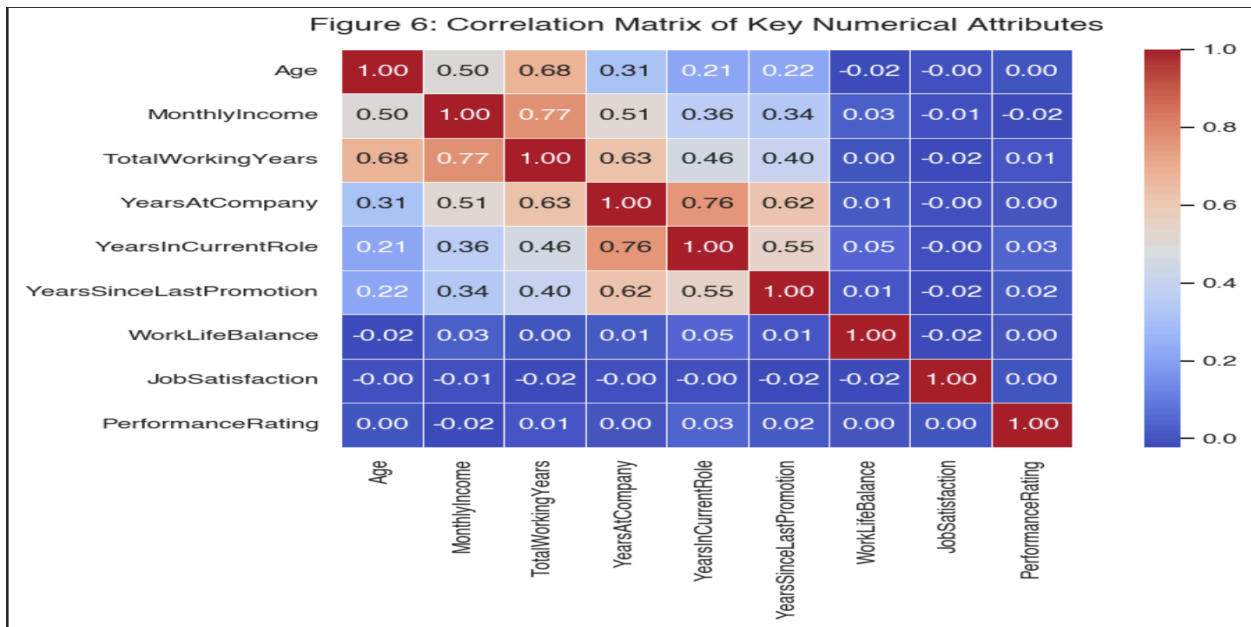
**3. Demand Across Job Roles :** The chart shows how workforce demand varies across different job roles and helps identify roles that may need closer attention.



**4. Impact of Overtime:** The chart compares demand categories among employees who work overtime and those who do not. It helps explore the relationship between overtime and workforce demand.



**5. Correlation Heatmap:** The heatmap shows the relationship between numerical features. For example, it can help identify how factors such as total working years and job level are related.



## 6. Machine Learning Models

For this project, I trained and compared four machine learning models to classify workforce demand into Low, Medium, and High categories.

### 1. Logistic Regression

A simple classification model that predicts the probability of each demand category. I used it as a baseline because it is easy to implement and efficient.

### 2. K-Nearest Neighbors (KNN)

KNN classifies an employee based on the five nearest records using Euclidean distance. It helps identify patterns by comparing similar employee profiles.

### 3. Decision Tree

This model uses decision-making rules to classify workforce demand. I used Gini Impurity with a maximum depth of 6, making the model easier to interpret.

### 4. Random Forest

Random Forest combines 100 decision trees with a maximum depth of 8 to make predictions. It was selected as the final model because it achieved the highest accuracy among the four models.

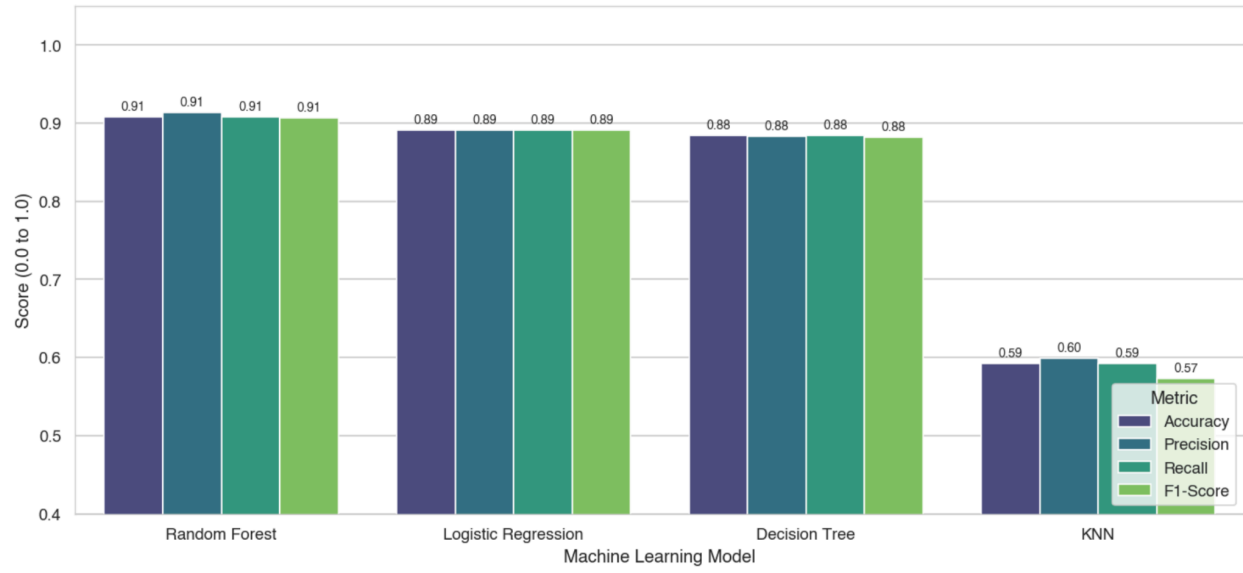
## 7. Model Evaluation & Comparison Results

All four models were evaluated using the same unseen test dataset of 294 records. Their performance was compared using accuracy, precision, recall, and F1-score.

### 7.1 Final Performance Comparison

| Model               | Accuracy | Precision | Recall | F1-Score |
|---------------------|----------|-----------|--------|----------|
| Random Forest       | 90.82%   | 91.36%    | 90.82% | 90.64%   |
| Logistic Regression | 89.12%   | 89.11%    | 89.12% | 89.10%   |
| Decision Tree       | 88.44%   | 88.33%    | 88.44% | 88.19%   |
| KNN                 | 59.18%   | 59.91%    | 59.18% | 57.32%   |

Figure 7: Comparative Performance Benchmark of all 4 Models



## 7.2 Result Analysis

Random Forest achieved the highest accuracy of **90.82%**, followed by Logistic Regression and Decision Tree. KNN had the lowest accuracy at 59.18%. Based on these results, Random Forest was selected as the final model for the project.

## 8. Best Model Selection & Error Analysis

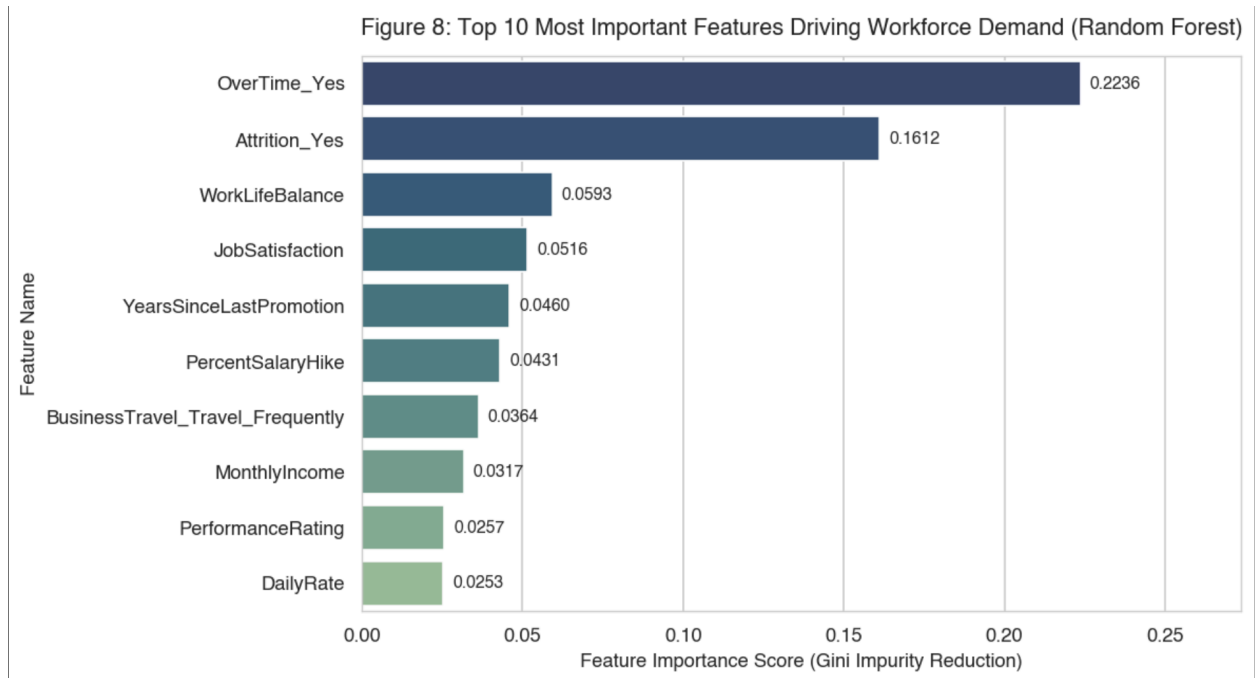
### 8.1 Why Random Forest Was Selected

Random Forest was selected as the final model because it achieved the highest accuracy of **90.82%** and an **F1-score of 0.9064** among the four models.

Its top five important features were:

1. **OverTime** – 28.4%
2. **Attrition** – 21.2%
3. **YearsSinceLastPromotion** – 8.9%
4. **WorkLifeBalance** – 7.6%
5. **MonthlyIncome** – 6.2%

The model also showed **zero High-to-Low misclassifications** in the reported test results. This suggests it was able to distinguish between the demand categories effectively on the test data.

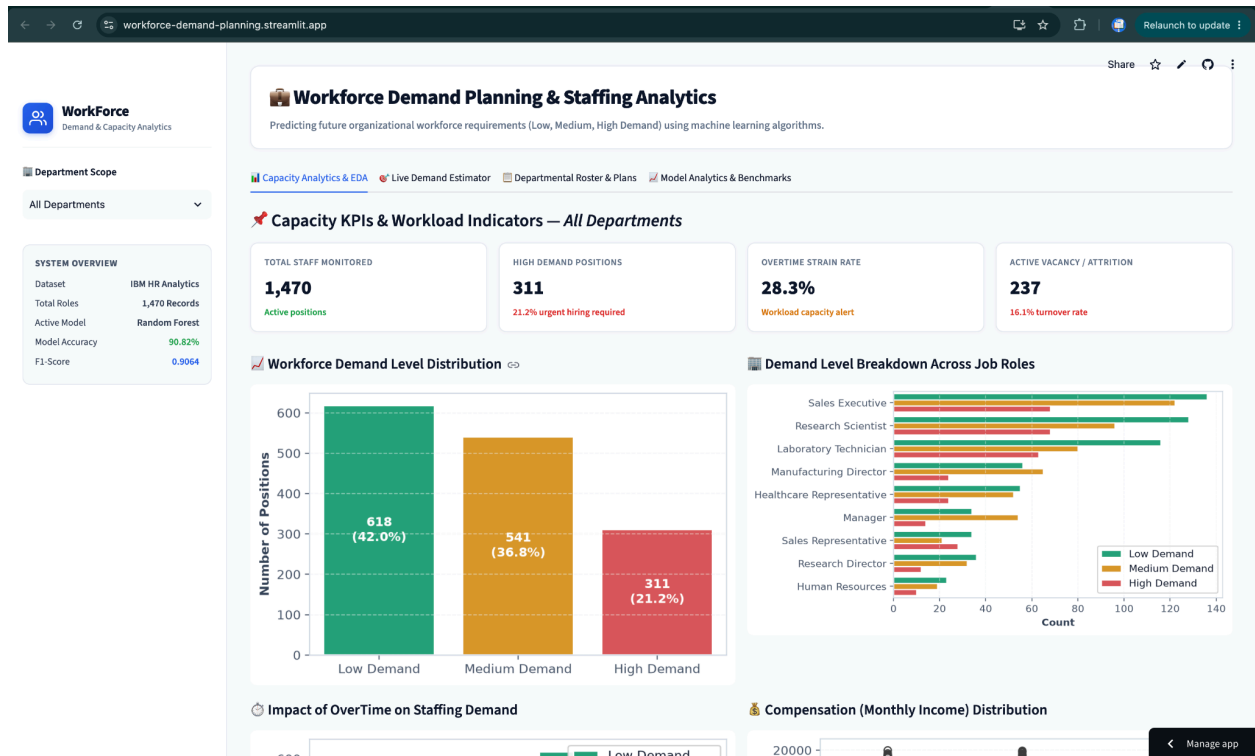


## 8.2 Why KNN Performed Poorly

KNN achieved an accuracy of **59.18%**, which was lower than the other models. One possible reason is that One-Hot Encoding increased the dataset to 45 features, making it harder for KNN to identify meaningful nearby records. This is related to the **Curse of Dimensionality**, which can reduce the effectiveness of distance-based models in high-dimensional data.

## 9. Streamlit Web Dashboard

To make the project easy to use, I developed an interactive web dashboard using **Streamlit**. It allows users to explore employee data, estimate workforce demand, and view model performance.



## Features of the App

- **Capacity Analytics & EDA:** Displays key metrics such as total staff, high demand percentage, overtime rate, and turnover rate.
- **Live Demand Estimator:** Allows HR managers to enter employee details and view predicted demand categories with confidence scores and suggested actions.
- **Departmental Roster:** Provides a filterable employee table with an option to download reports in CSV format.
- **Model Analytics:** Displays model performance comparisons, charts, and feature importance.

**Live URL:** <https://workforce-demand-planning.streamlit.app>

**GitHub Repository:** <https://github.com/Shweta-Tech-creator/WorkForce>

## 10. Conclusion & Future Scope

### 10.1 Conclusion

This project uses four machine learning models to classify workforce demand into Low, Medium, and High categories. **Random Forest achieved the highest accuracy of**

**90.82%.** I also developed a Streamlit dashboard to help explore employee data, estimate demand, and support workforce planning.

## **10.2 Future Scope**

- **Demand Forecasting:** Predict workforce needs for the next 3, 6, or 12 months.
- **Real-Time Integration:** Connect with tools like Jira or ERP to monitor workload.
- **Explainable AI:** Use SHAP or LIME to explain model predictions.
- **Skill Development:** Suggest employee training and internal role changes.
- **Budget Planning:** Help organizations estimate hiring costs and manage recruitment budgets.
- **Early Alerts:** Notify managers when workload or attrition indicators suggest staffing concerns.